UC BERKELEY EECS · RESEARCH OVERVIEW

Building and evaluating agents for the real world.



Yuanbo Pang · Berkeley RDI/BAIR (Prof. Dawn Song) · Tomizuka Lab · Equile (independent project)

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IN ONE SLIDE

Three threads, one question: can we trust what agents do?

Evaluate agents

Agents' Last Exam — 1K+ executable tasks, automated graders, verifiable rewards. Co-first author.

Evaluate data

roboqa — what robot-learning data is worth, measured at EgoDex scale. Tomizuka Lab.

Run agents as infrastructure

Equile — reproducible computing environments with provenance for every result.

Output

2 preprints (HF #1 of week / #2 of day) + 1 under review at ICLR 2027.

AGENTS' LAST EXAM · ARXIV:2606.05405 · UNDER REVIEW, NEURIPS 2026

Measuring frontier agents on work that matters.

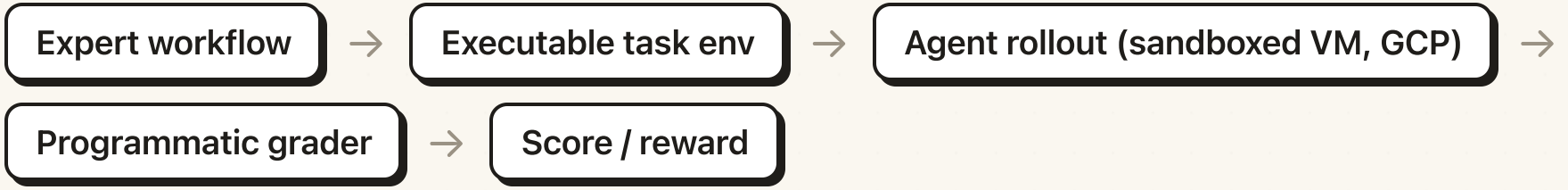
- **1K+ expert-designed, long-horizon tasks** across 55 subdomains — real economic work, not puzzles.
- Every task is an **executable environment**: `load / start / evaluate`, sandboxed VM, computer-use included.
- **Programmatic graders** — no hand labels in the scoring loop; scores double as **verifiable rewards (RLVR)**.

1,000+	55	#1	310
executable tasks	subdomains	HF paper of the week	authors — I'm co-first

My role: led the task pipeline and auto-scoring; run evals with frontier model teams.

ALE · UNDER THE HOOD

From expert workflow to trustworthy number.



- Evaluator families: **hash · tabular · geometric · visual · behavioral** — picked per task, checked against drift.
- Trajectory analysis: find *why* an agent failed, report failure modes back to the labs (OpenAI, Anthropic, Meta TBD, Qwen, Seed).
- V2 (leading): **3D/spatial environments** (Blender, Isaac Lab) and **ultra-long-horizon team-of-agents tasks** from production software.

ROBOQA · TOMIZUKA LAB, ADVISED BY ZHUO XU

What is a robot dataset actually worth?

- Three dimensions, **never averaged**: recording health · distribution diversity · trainability.
- **Agent workflows inside the pipeline**: VLM-judged annotation audits; raw captures auto-converted to a canonical, embodiment-tagged format.
- **Train + post-train VLAs on 500 h** of curated data; score the resulting policies in simulation.
- Every score published with provenance: data fingerprint, measure versions, judge, commit.

338K
episodes (EgoDex)

829 h
of egocentric data

500 h
VLA train + post-train

ALSO ON THE BENCH

Environments are the common thread.

ScienceIDE HF #2 of the day

Turning the world's scientific codebases into agent-learnable environments; trained PhAI-IDE model family on verified trajectories. arXiv:2609.19134.

Agent's GameDev Exam ICLR 2027 · under review

Agent evaluation in game-development environments — with Dawn Song's group.

Equile — "the dry lab for AI scientists"

Independent project, Apr–Oct 2026. Reproducible computing environments + clusters for research agents; every result traceable to how it was made.

WHERE THIS IS GOING

The questions I want to spend years on.

1. Can we reliably evaluate long-horizon agents — evaluators whose verifiable rewards are strong enough to both **measure** frontier agents and **train** them?

2. What data do robots actually need? Can egocentric human demonstrations become as useful to robot learning as web text is to language models?

3. What does it take to run agents as real infrastructure — sandboxed, reproducible, schedulable — and how does that change how we study them?

THANK YOU

Let's talk.

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ALE: agents-last-exam.org · arXiv:2606.05405 / ScienceIDE: arXiv:2609.19134